Task 1: Guided Activity, Restaurant bill calculator

```
import Foundation

let itemPrice: Double = 450.0
let quantity: Int = 3

let subtotal = itemPrice * Double(quantity)
print("Subtotal: ₹\(subtotal)")

let hasCoupon = true

let discount = (hasCoupon && subtotal > 1000) ? subtotal * 0.10 : 0.0
print("Discount: ₹\(discount)")

let discountedAmount = subtotal - discount
let gst = discountedAmount * 0.18
let total = discountedAmount + gst

print("GST: ₹\(gst)")
print("Total Amount: ₹\(total)")

let people = 4
let share = total / Double(people)

print("Each Person's Share: ₹\(share)")

let amount = 1234
let hundreds = amount / 100
let remainder = amount % 100

print("Amount: ₹\(amount)")
print("Hundreds: \(hundreds)")
print("Remainder: ₹\(remainder)")
```

OUTPUT:

```
Subtotal: ₹1350.0
Discount: ₹135.0
GST: ₹218.7
Total Amount: ₹1433.7
Each Person's Share: ₹358.425
Amount: ₹1234
Hundreds: 12
Remainder: ₹34
```

Task 2: Practice Task, Leap year and range checks

```
import Foundation

let year1 = 1900
let year2 = 2000
let year3 = 2024
let year4 = 2026

let leap1 = (year1 % 4 == 0) && ((year1 % 100 != 0) || (year1 % 400 == 0))
let leap2 = (year2 % 4 == 0) && ((year2 % 100 != 0) || (year2 % 400 == 0))
let leap3 = (year3 % 4 == 0) && ((year3 % 100 != 0) || (year3 % 400 == 0))
let leap4 = (year4 % 4 == 0) && ((year4 % 100 != 0) || (year4 % 400 == 0))

print("1900 is a leap year: \(leap1)")
print("2000 is a leap year: \(leap2)")
print("2024 is a leap year: \(leap3)")
print("2026 is a leap year: \(leap4)")

let number = 75
let isInRange = (1...100).contains(number)

print("Number is between 1 and 100: \(isInRange)")

let num = 24
let isEvenAndNotMultipleOf10 = (num % 2 == 0) && (num % 10 != 0)

print("Even and not a multiple of 10: \(isEvenAndNotMultipleOf10)")

let nickname: String? = nil
let displayName = nickname ?? "Guest"

print("Nickname: \(displayName)")
```

OUTPUT:

```
1900 is a leap year: false
2000 is a leap year: true
2024 is a leap year: true
2026 is a leap year: false
Number is between 1 and 100: true
Even and not a multiple of 10: true
Nickname: Guest
```

Task 3: Predict Task, What does it print?

```
print(7 / 2)

print(7 % 2)

print(-7 % 3)

print(7.0 / 2)

print(10 - 2 * 3 + 4 / 2)

print(5 > 3 && 2 > 4 || true)

print(1 << 3)

print(6 & 3, 6 | 3, 6 ^ 3)

let x: UInt8 = 255

print(x &+ 1)
```

OUTPUT:

```
3
1
-1
3.5
6
true
8
2 7 5
0
```